Support Vector Machines Used to Estimate the Battery State of Charge

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***Abstract*—The aim of this study is to estimate the state of charge (SOC) of a high-capacity lithium iron manganese phosphate ($LiFeMnPO_4$) battery cell from an experimental dataset using a support vector machine (SVM) approach. SVM is a type of learning machine based on statistical learning theory. Many applications require accurate measurement of battery SOC in order to give users an indication of available runtime. It is particularly important for electric vehicles or portable devices. In this paper, the proposed SOC estimator extracts model parameters from battery charging/discharging testing cycles, using cell current, cell voltage, and cell temperature as independent variables. Tests are carried out on a 60 Ah lithium-ion cell with the dynamic stress test cycle to set up the SVM model. The SVM SOC estimator maintains a high level of accuracy, better than 6% over all ranges of operation, whether the battery is charged/discharged at constant current or it is operating in a variable current profile.**

***Index Terms*—Lithium batteries, modeling, state of charge (SOC), support vector machines (SVMs), support vector regression (SVR).**

I. Introduction

LITHIUM-ION is the newest battery chemistry, well understood, and widely used in numerous applications thanks to its high cell voltage, high energy density, long lifetime, and exceptional cyclability. One of the most important battery parameters is the battery state of charge (SOC). Knowing the amount of energy left in a battery compared with the energy it had when it was full gives the user an indication of how much longer a battery will continue to perform before it needs recharging. Using the analogy of a fuel tank in a car, SOC estimation is often called the "Gas Gauge" or "Fuel Gauge" function. The SOC is defined as the available battery capacity expressed as a percentage of its rated capacity, where 100% implies a full battery state and 0% the empty state. The battery capacity is usually measured in Ampere-hour. Determining and controlling the SOC is critically important in applications for electric vehicle (EV), hybrid electric vehicle (HEV), or portable devices. These applications require accurate measurement of battery SOC in order to give users an indication of available runtime. The SOC estimation can also be useful to avoid detrimental situations, such as overdischarging and overcharging, which can lead to a reduction in the battery life. An accurate SOC indication is thus important for user convenience and for prolonging the lifetime of batteries. However, many examples of poor accuracy and reliability are found in practice. The main problem of designing an accurate SOC indication system is the unpredictability of both the battery and user behavior.

Several techniques have been reported for measuring or estimating the SOC of battery cells [1]. The most common methods include current integration techniques, artificial neural networks (ANNs) and fuzzy logic-based estimations [2], Kalman filter-based estimators [3]–[5], and others [6]–[9]. Most of these methods have been widely used and have achieved acceptable results in different applications. One of the most common technique for calculating SOC is the ampere-hour counting. The method is based on current measurement and integration during the charge and discharge states. One drawback of current integration techniques is their reduced accuracy. If the SOC algorithm is based on current measurement and integration, the error caused by the current measurement inaccuracy will accumulate over time. For many SOC levels, this estimator carries an average $\pm 15\%$ error. Other methods based on neural networks or Kalman filter-based estimators exhibit greater performance, but a high implementation cost is required due to their computational complexity. For example, extended Kalman filter (EKF) approach [3] maintains accuracy about $\pm 5\%$. Nevertheless, the EKF SOC estimator is a high computational complexity technique that requires a high implementation cost.

Support vector machine (SVM) techniques [10]–[12] have been developed greatly in recent years, giving rise to a new kind of learning machines that use the central concept of the SVM approach, namely, kernels, for a number of learning tasks. Kernel machines provide a modular framework that can be adapted to different tasks and domains by using different kernel functions (i.e., linear, polynomial, sigmoid, or radial basis) and the base algorithm [13]. They are used in a variety of fields, including biomedicine and bioinformatics [14], image analysis and artificial vision [15], and other engineering fields [16], [17],

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TABLE I
HIGH-CAPACITY $LiFeMnPO_4$ BATTERY CELL CHARACTERISTICS

Parameter	Value
Manufacturer Model Nominal capacity	GBS Energy GBS LFMP60AH 60 Ah
Nominal voltage	3.2 V
Charging/dischar. cut-off voltage	3.55 V / 2.8 V
Recommend charging current Maximum discharge current (t<10s) Maximum continuous discharge curr.	0.25 C 15 A 600 A 180 A
Cycle life (at 80% DoD, 0.3 C)	3000 times
Operating temp. (charge/discharge)	0ºC to 45ºC / −26ºC to 65 ºC
Weight	2 kg
Size (length/width/height)	126/65/180 mm

including the battery modeling [18]. Like conventional feedforward neural networks, such as the multilayer perceptron [19], SVM has been used by researchers to solve classification and regression problems [20]. SVM techniques are based on the structural risk minimization principle [21]. This principle has shown to be superior to the traditional principle of empirical risk minimization employed by conventional neural networks. While neural networks may have local minimization problems, the learning algorithm of SVMs is to solve a quadratic programming, which has the global optimal solution. This feature gives SVM greater generalization ability. However, it is not readily apparent that SVMs are meant to be a total replacement for ANN. For instance, multilayer feedforward networks, such as MLPs, are in direct competition with SVMs. One specific benefit that these models have over SVMs is that their size is fixed [11]. This means that the SVM model size could increase much more. Other advantage of artificial neural networks over SVMs is that artificial neural networks may have any number of outputs, while SVMs have only one. SVM is also used as an alternative to the traditional regression approaches because it is more robust than a least-squares estimator.

In this research study, SVM for regression (SVR) is used as an automated learning tool with a different focus to successfully predict the SOC of a high-capacity lithium iron manganese phosphate (LiFeMnPO4) battery cell as a function of cell voltage, cell current, and cell temperature.

II. MATERIALS AND METHODS

A. Battery Cell and Test Workbench

A $LiFeMnPO_4$ battery cell with a nominal capacity of 60 Ah is used in this study. This cell is an ideal energy source not only for electric bikes, scooters, HEVs, and EVs, but also for energy storage devices used in solar and wind electricity generation applications and other multipower systems in the military use. The main cell characteristics are summarized in Table I.

The battery-testing platform consists of programmable stand-alone instruments and a computer-control unit using LabVIEW software. The test workbench block diagram is shown in Fig. 1. Workbench uses stand-alone instruments for charging and discharging purposes. Stand-alone instruments are modular/independent instruments that support a wide variety of functions in the power and measurement field. In this case, a programmable power source (Sorensen DHP 60–220 M1-M9D, 0–60 V, 0–220 A) is used for cell charging.

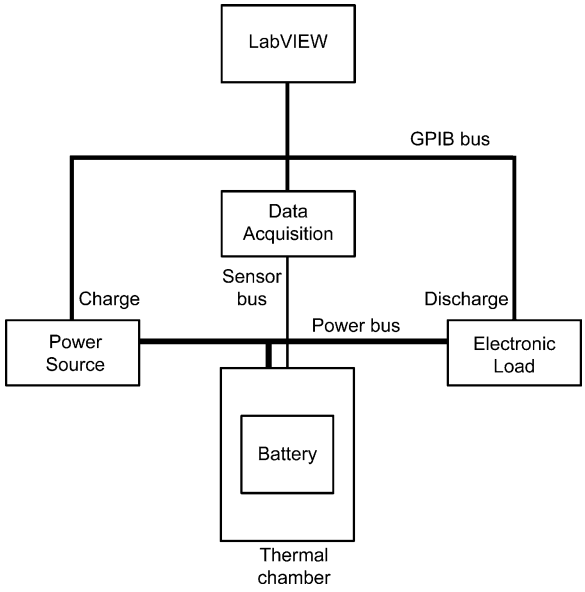


Fig. 1. General architecture of the battery test workbench.

An electronic load (Agilent 6050A with three 60504B modules) is used for discharging purpose. The battery voltage, temperature, and other variables used for charging sensing termination [22] are monitored using a modular data acquisition system (Agilent 34970A) with a total accuracy of 260 μV for dc measures. Battery cell temperature is measured using an RTD attached to the center of the module. Internal power source and electronic load meters are used to provide current measurements with an accuracy of 0.12% $\pm$ 130 mA. The battery performance largely depends on temperature, so all tests start at the initial temperature of 23 °C $\pm$ 2 °C in an electronically controlled environmental chamber. The workbench is controlled by a custom software application written in the National Instrument LabVIEW software. A description of the workbench and operating details can be found in [23].

B. Experimental Dataset

The most reliable test for determining the SOC of a battery is a discharge test at a particular C-rate under controlled conditions. This test includes a consecutive battery recharge. Charging and discharging currents are generally expressed as multiples of C. C-rate is a value that expresses the rated current capacity of a cell or battery. For example, a cell discharging at the C-rate will deliver its nominal rated capacity for 1 h. The time to discharge a battery is inversely proportional to the discharge rate.

In this paper, three consecutive cycles of charging at 0.3 C (18 A) and discharging at 0.33 C-rates (20 A) are performed using a GBS Energy $LiFeMnPO_4$ battery cell with a nominal capacity of 60 Ah. These initial cycles are used for cell

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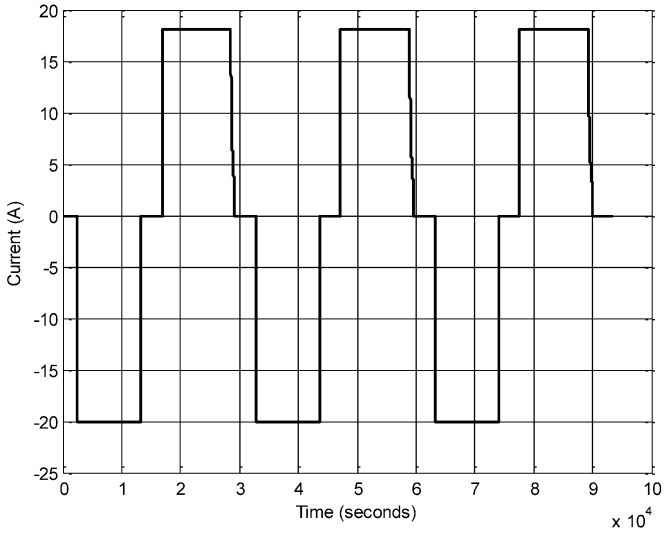


Fig. 2. Initial charging and discharging cycles used for battery conditioning.

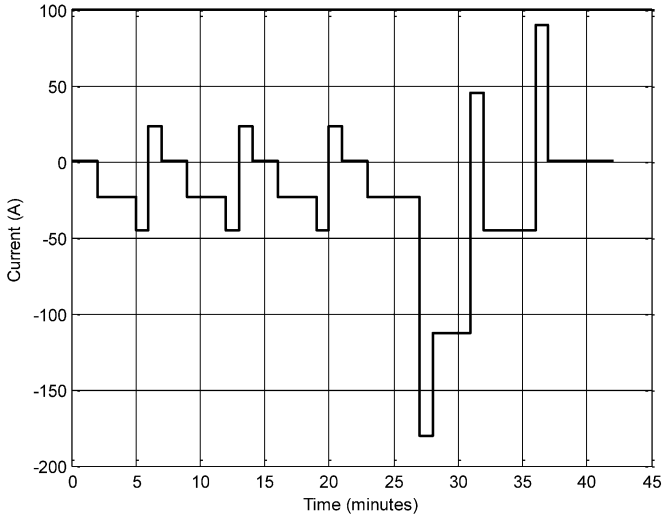


Fig. 3. DST profile.

conditioning and for taking SOC reference measurements. Fig. 2 shows the initial charging and discharging cycles, where a negative current value means a discharge current. It must be noted that the charging/discharging profile shown in Fig. 2 represents a process that takes hours. Initial battery conditioning is performed in two steps using the workbench described previously. During charging step, the battery is fully charged using the usual constant-current-constant-voltage (CCCV) charging method at a 0.3 C-rate charging current until reaching 3.6 V. In the constant voltage (CV) mode, the voltage is then kept constant at 3.6 V while the charging process continues in the CV mode until the charge current has dropped to 0.05 C (3 A). The full SOC (100%) is reached in this situation. Second, after a rest period, the battery is discharged at constant current (0.33 C) until the cutoff voltage is reached. This process is repeated three times. This particular conditioning test is used to provide data to the SVM model under "static" (constant current) or "slow change" conditions. The SVM SOC estimator should maintain a high level of accuracy whether the system is changing slowly or it is operating in a dynamic way. Therefore, after this initial cycling, a dynamic stress test (DST) according to USABC is used [24]. This variable power discharge regime can effectively simulate a driving cycle testing and can be implemented with equipment at most test laboratories and developers. Fig. 3 shows a graphical representation of the DST test profile used in this paper. Table II tabulates the corresponding DST values. DST steps are scaled in time and the current is a percentage of the maximum continuous discharge current of the cell (180 A). A negative current value in Fig. 3 means a discharge current. It must be noted that the x-axis in Fig. 3 represents time in minutes. DST test is used to provide a "dynamic" charging–discharging data to the SVM model.

The main goal of this research work is to obtain a model to estimate the SOC using the SVM technique. Cell variables, such as the voltage, current, and temperature, are measured during the battery conditioning test and DST test. These data are used to identify the model parameters. Figs. 4 and 5 show the experimental data used for training the SVM model. Furthermore, the previous figures include the corresponding battery SOC to be predicted by the SVM model. The model training data cover operation from 100% SOC to 0% SOC and back up to 100% SOC in Fig. 4 (static dataset). All data points in this SOC range are used for training. The training current is ranging from 90 to −180 A in the DST test (dynamic dataset) (see Fig. 5). The SOC range goes from 100% to 0% in the DST test. Several DST cycles are considered until the cell reaches the cutoff voltage (2.8 V), as shown in Fig. 5. Additionally, several plateaus with 0-A current and steady-state SOC are also included as training data. It must be noted that the experimental SOC data shown in Figs. 4 and 5 were calculated using the internal current meters

TABLE II
LISTING OF DST PROFILE

Step No.	Duration (minutes)	Amplitude [%]	Current [A]
1	2	0.0%	0
2	3	−12.5%	−23
3	1	−25.0%	−45
4	1	12.5%	23
5	2	0.0%	0
6	3	−12.5%	−23
7	1	−25%	−45
8	1	12.5%	23
9	2	0.0%	0
10	3	−12.5%	−23
11	1	−25%	−45
12	1	12.5%	23
13	2	0.0%	0
14	4	−12.5%	−23
15	1	−100%	−180
16	3	−62.5%	−113
17	1	25%	45
18	4	−25%	−45
19	1	50%	90
20	5	0.0%	0

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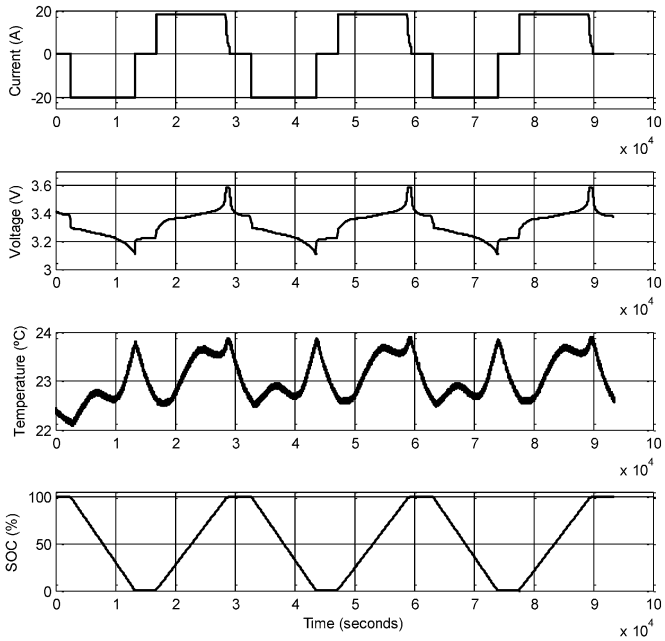


Fig. 4. Experimental dataset (static test).

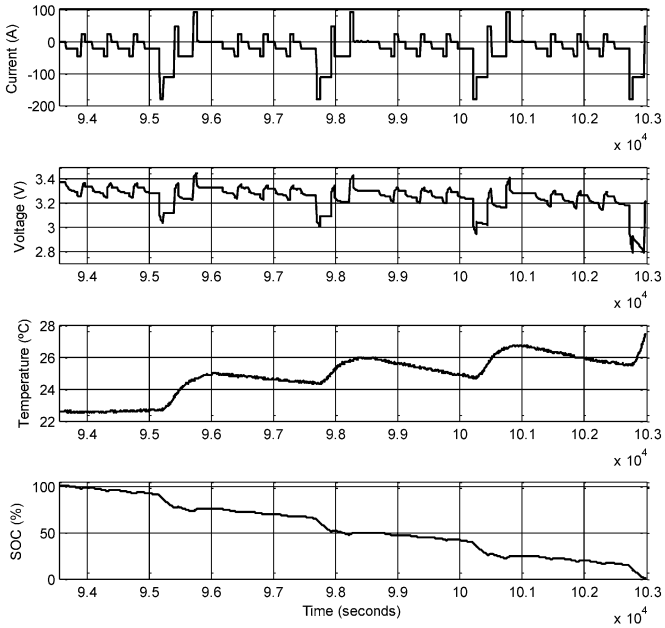


Fig. 5. Experimental dataset (DST test).

provided by the workbench electronic load (discharging) and the power supply (charging).

C. SVM Method

SVMs are a set of related supervised learning methods used for classification and regression that can universally approximate any multivariate function to any level of accuracy [11]. SVMs were originally developed to solve classification problems. They were later generalized to solve regression problems in a method called SVR [10], [25]. The SVM goal is then to find a functional form that can correctly predict new cases not tackled before. This can be achieved by training the SVM model on a sample set called training set, a process that involves the sequential optimization of an error function. Depending on the definition of this error function, two types of SVM models can be recognized [11], [13], [20], [25]: 1) $\varepsilon-$SVM regression; and 2) $\nu-$SVM regression. This research problem is quadratic with linear constraints, and so, the Karush–Kuhn–Tucker optimality conditions are necessary and sufficient [11], [20]. The solution, which can be obtained from the dual problem, is a linear combination of a subset of sample points denominated support vectors [16], [17], [20].

TABLE III
SET OF CELL INPUT VARIABLES

Input variables	Variable name
Cell voltage (V)	Voltage
Cell current (A)	Current
Cell temperature (°C)	Temp

Many regression problems cannot be linearly regressed in the space of the inputs. However, they might be regressed in a higher dimensional feature space given a suitable mapping (the kernel trick) [20]. Different kernel functions are described in [11], [12], and [25] such as radial basis function (RBF) kernel, polynomial kernel, and so on. The RBF is by far the most popular choice of kernel types used in SVMs. This is mainly because of their localized and finite responses across the entire range of the real x-axis. To sum up, to use an SVM to solve a regression problem for data that is not linearly separable, a kernel must be chosen. Then, the relevant parameters that can be expected to map the nonlinearly separable data into a feature space where it is linearly separable must be selected. Furthermore, it is well known that the SVM techniques are strongly dependent on the SVM hyperparameters (C, ν, γ), [11], [20], [26]: the regularization factor C, the hyperparameter ν that defines the SVM type regression, and γ that represents the kernel parameter if an RBF is chosen. In this paper, each combination of hyperparameter choices is checked using cross validation, and the parameters with best cross-validation accuracy are chosen.

III. ANALYSIS OF RESULTS AND DISCUSSION

The cell input variables considered in this study are shown in Table III. The battery voltage, temperature, and current measurements are given as input variables to the SVM model. The model is trained using the dataset shown in Figs. 4 and 5. An exhaustive cross-validation algorithm is used to guarantee the predictive ability of the SVM model. In this study, 10-fold cross validation is used [26]. Cross validation is a way to predict the fit of a model to a hypothetical validation set when an explicit validation set is not available.

One round of cross validation involves partition, the dataset is randomly divided into l disjoint subsets of equal size, and each subset is used once as a validation set, while the other $l-1$ subsets are put together to form a training set. In the simplest

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TABLE IV
OPTIMAL HYPERPARAMETERS (SVM MODEL)

Input variables	Variable name
SVM-type	$\nu-$regression
SVM-kernel	Radial basis function (RBF)
C	1
γ	0.125
ν	1
Num. of support vectors	903

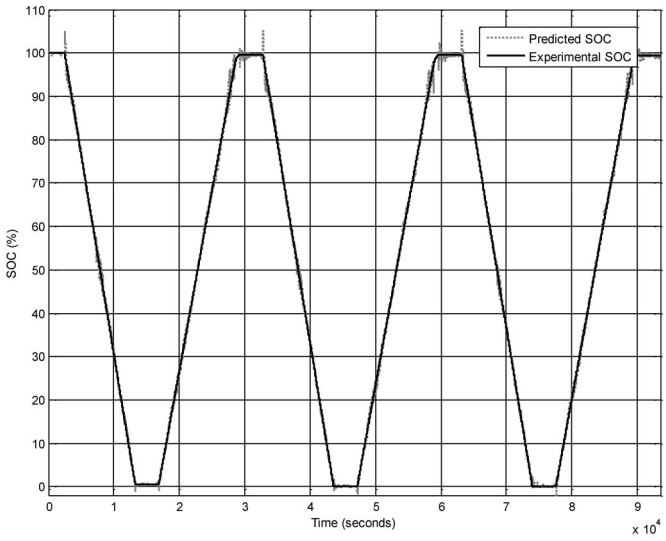


Fig. 6. Predicted SOC versus experimental SOC (static test).

case, the average accuracy of the l validation sets is used as an estimator for the accuracy of the method. In this way, 10-fold cross validation was used. The combination of the hyperparameters with the best performance is also chosen using 10-fold cross validation [13], [16], [20]. Table IV shows the optimal hyperparameters of the fitted SVM model used in this research work.

The comparison between predicted SOC values and experimental SOC data is shown in Figs. 6 and 7. Fig. 6 shows the experimental SOC versus predicted SOC using the SVM model for the static dataset (constant charging/discharging profile). Fig 7 shows the corresponding results for the DST dataset. The results clearly show that the fitting is significantly good. Moreover, Figs. 6 and 7 show that there is no drift over time in SOC estimation using the SVM model. Small peaks can be seen in Fig. 6 at the points that match the input current transient points, see a zoomed-in view in Fig 8. Note that the input current is constant except at the step-change points, where current jumps from one steady state value to another (see Fig. 2). From the mathematical point of view, these peaks correspond to points of discontinuity of the current function and reflect the peculiar manner in which SVM behave at an abrupt change. However, these peaks could be corrected using outliers cancellation. The peaks (outliers) appear with an SOC value greater than 100% and lower than 0% and then can be easily identified and canceled. These outliers represent a moderate problem in relation with the global model accuracy. The coefficient of determination was chosen in this research study to estimate the goodness of fit of the SVM model [27]. This ratio indicates the proportion of total variation in the dependent variable explained by the model.

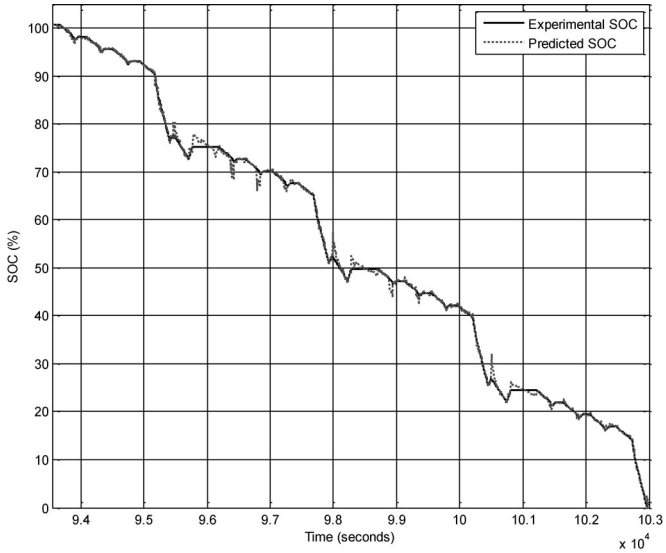


Fig. 7. Predicted SOC versus experimental SOC (DST test).

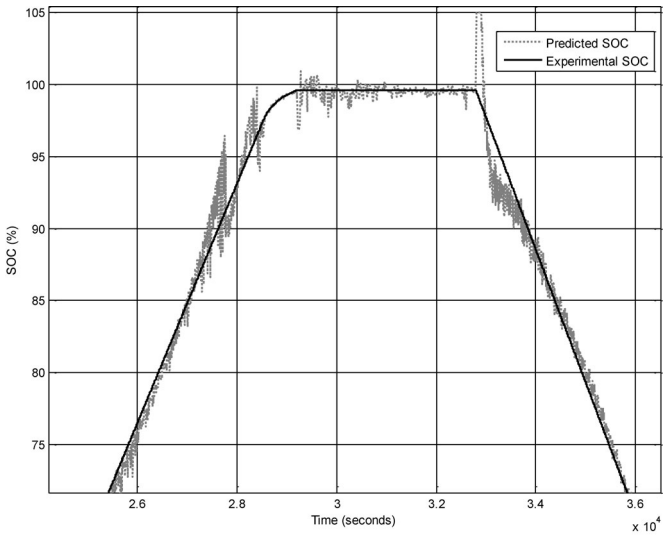


Fig. 8. Zoomed-in view of SOC prediction for the static test.

A dataset takes values t_i, each of which has an associated modeled value y_i. The former are called the observed values and the latter are often referred to as the predicted values. Variability in the dataset is measured through different sums of squares.

1) $SS_{\mathrm{tot}} = \sum_{i=1}^{n} (t_i - \bar{t})^2$ is the total sum of squares, proportional to the sample variance.

2) $SS_{\mathrm{reg}} = \sum_{i=1}^{n} (y_i - \bar{t})^2$ is the regression sum of squares, also called the explained sum of squares.

3) $SS_{\mathrm{err}} = \sum_{i=1}^{n} (t_i - y_i)^2$ is the residual sum of squares.

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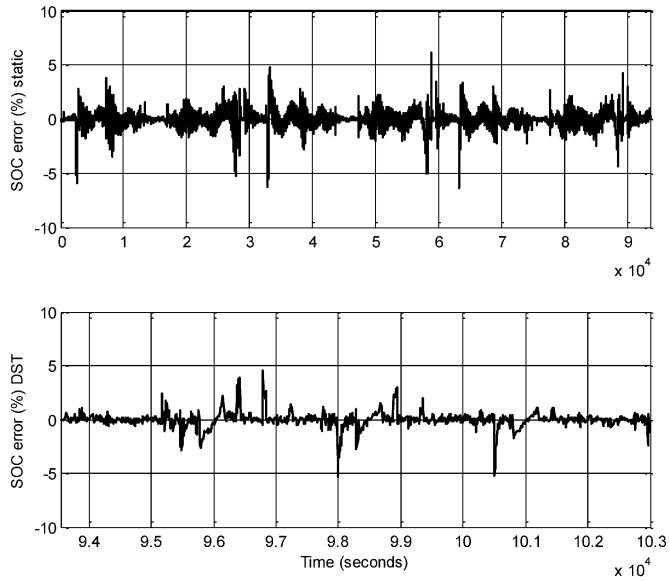


Fig. 9. SOC prediction errors for the static dataset (upper) and DST dataset (lower).

In these sums, $\bar{t}$ is the mean of the n observed data

$$\bar{t} = \frac{1}{n}\sum_{i=1}^{n} t_i.$$

Bearing in mind the aforementioned sums, the general definition of the coefficient of determination is

$$R^2 \equiv 1 - \frac{SS_{\rm err}}{SS_{\rm tot}}.$$

A coefficient of determination value of 1.0 indicates that the regression curve fits the data perfectly. In this study, the fitted SVM model has a coefficient of determination equal to 0.98. This result confirms the excellent accuracy for the SOC prediction using the SVM technique. This high goodness of fit is related to the selected input variables and the type of kernel chosen for the model. In this research paper, an SVM regression technique with an RBF kernel is selected. This particular selection of kernel yields to better results in this kind of application than other kernels [18] because, in practice, RBF kernels show a better performance in high nonlinear problems [11].

Two additionally evaluation criteria were adopted for this study to compare the model performance: the error between experimental and predicted SOC and the root-mean-square error (RMSE). Fig. 9 shows a detailed view of errors between the experimental and predicted SOC. In this way, prediction errors are kept below 6% in both tests (static and dynamic). Finally, the RMSE is 0.71% over the whole test.

To test the ability of the modeling technique, the same SVM method is applied to predict the cell voltage using the same high-capacity $LiFeMnPO_4$ battery cell. Table V shows the model input variables used to predict the cell voltage. The voltage predicted by the SVM model and the experimental voltage using DST data is shown in Fig. 10. The fitted voltage has a coefficient of determination equal to 0.97. The relative errors between experimental voltage and predicted voltage are expressed in percentage in Fig. 11.

TABLE V
SET OF CELL INPUT VARIABLES FOR VOLTAGE PREDICTION

Input variables	Variable name
Cell SOC (%)	SOC
Cell current (A)	Current
Cell temperature (ºC)	Temp

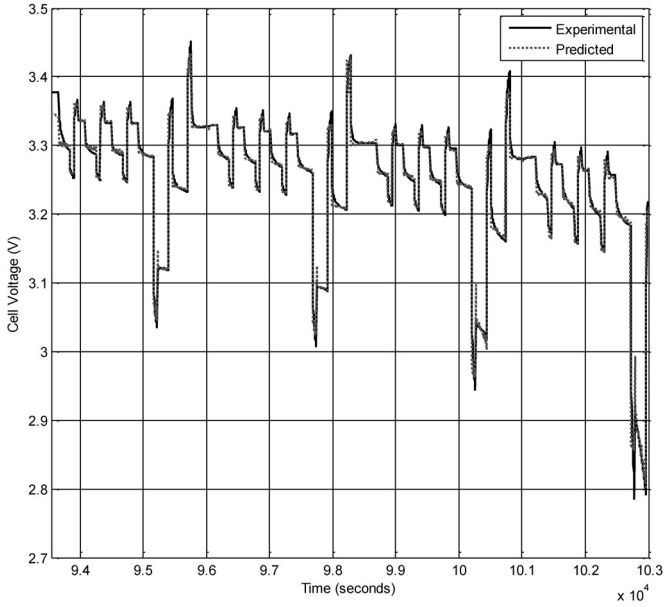


Fig. 10. Predicted and experimental cell voltages using the DST dataset.

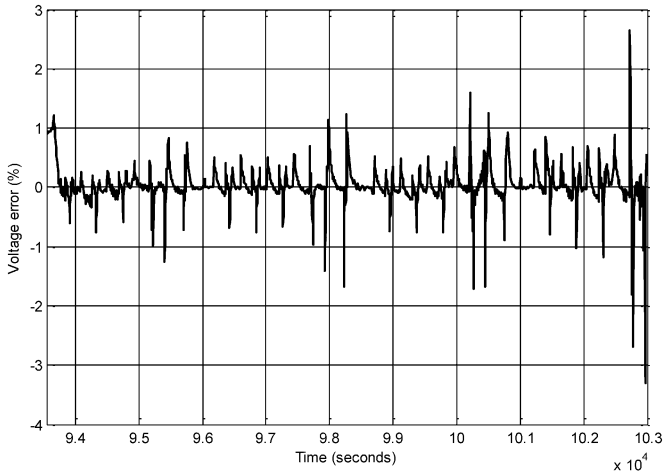


Fig. 11. Relative errors between the predicted and experimental cell voltages (expressed in percentage).

This result also confirms an excellent goodness of fit in the prediction of cell voltage in dynamic conditions. In this case, the relative maximum error is kept below 4% and the RMSE is 0.4%.

It is important to know the implementation requirements of SVM in a practical application. In this way, the SVM model can make predictions quickly due to the fact that the prediction

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function simply has to evaluate a manageable number of support vectors. This result means a minimal computation load. For example, SVM usually does not require matrix inversion and intensive calculations that are required by others approaches [3]. Using the right training data and the proper kernel functions, the SVM predictive model calculates SOC in a reduced time with minimum storage requirements. All these features make the SVM model suitable for implementation on a low-cost microcontroller-based battery management system (BMS) to fulfill the function of the SOC prediction. One of the main tasks of a BMS is to keep track of a battery's SOC. Finally, the model can be easily modified to successfully fit data from different batteries, because SVM models are more robust and flexible than classical linear regression models [20], [27].

IV. Conclusion

Determining the SOC is particularly important for EV, HEV, or portable devices. However, as a battery is a nonlinear system, it is difficult to establish the relationship between the SOC and measurable battery parameters. The main purpose of this research work was to build an SOC predictive model using the SVR approach for a high-capacity $LiFeMnPO_4$ battery cell. The proposed SOC estimator extracts model parameters from battery charging/discharging testing cycles, using voltage, current, and temperature data as independent variables. Tests are performed on a 60-Ah lithium-ion cell with the DST cycle to set up the SVM model.

The results confirm the accurate predictions of SOC: a coefficient of determination as high as 0.98 is obtained. Moreover, maximum errors are kept below 6% and the RMSE is 0.71% over the whole test. The SVM SOC estimator keeps this level of accuracy whether the battery is being fully charged/discharged at constant current or it is operating under a DST.

The developed model is able to predict the SOC quickly and fairly accurately using the right training data and propel kernel functions (RBF). A quick SOC prediction is also obtained, seeing as the prediction function simply has to evaluate a manageable number of support vectors, which can be manipulated by an inexpensive digital processor. The SVM model may thus be implemented on a low-cost microcontroller-based BMS system to fulfill the function of the SOC prediction. It might offer accuracy comparable with more sophisticated techniques at lower computational cost.

The SVM model is tested in a similar way to predict the cell voltage, obtaining accurate predictions under dynamic conditions with a coefficient of determination of 0.97. Maximum voltage errors are kept below 4% and the RMSE is 0.4% over the whole test.

Finally, the authors of this research work have confidence that the results obtained will be useful to tackle new future studies in other similar battery cells by applying the same methodology.

References

[1] P. Sabine, P. Marion, and J. Andreas, "Methods for state-of-charge determination and their applications," *J. Power Sources*, vol. 96, pp. 113–120, 2001.

[2] C. Chan, L. E. Lo, and S. Weixiang, "The available capacity computation model based on artificial neural network for lead-acid batteries in electric vehicles," *J. Power Sources*, vol. 87, pp. 201–204, 2000.

[3] G. Plett, "Extended Kalman filtering for battery management systems of LiPB-based HEV battery packs: Part 2. Modeling and identification," *J. Power Sources*, vol. 134, pp. 262–276, 2004.

[4] H. Dai, X. Wei, Z. Sun, J. Wang, and W. Gu, "Online cell SOC estimation of Li-ion battery packs using a dual time-scale Kalman filtering for EV applications," *Appl. Energy*, vol. 95, pp. 227–237, 2012.

[5] J. Kim, S. Lee, and B. H. Cho, "Complementary cooperation algorithm based on DEKF combined with pattern recognition for SOC/capacity estimation and SOH prediction," *IEEE Trans. Power Electron.*, vol. 27, no. 1, pp. 436–451, Jan. 2012.

[6] M. Verbrugge, "Adaptive, multi-parameter based state estimator with optimized time-weighting factors," *J. Appl. Electrochem.*, vol. 37, pp. 605–616, 2007.

[7] Y. Hu and S. Yurkovich, "Battery cell state-of-charge estimation using linear parameter varying system techniques," *J. Power Sources*, vol. 198, pp. 338–350, 2012.

[8] I. Kim, "Nonlinear state of charge estimator for hybrid electric vehicle battery," *IEEE Trans. Power Electron.*, vol. 23, no. 4, pp. 2027–2034, Jul. 2008.

[9] J. Kim, J. Shin, C. Chun, and B. H. Cho, "Stable configuration of a Li-ion series battery pack based on a screening process for improved voltage/SOC balancing," *IEEE Trans. Power Electron.*, vol. 27, no. 1, pp. 411–424, Jan. 2012.

[10] V. Vapnik, *Statistical Learning Theory*. New York, NY, USA: Wiley, 1998.

[11] I. Steinwart and A. Christmann, *Support Vector Machines*. New York, NY, USA: Springer, 2008.

[12] T. Hastie, R. Tibshirani, and J. H. Friedman, *The Elements of Statistical Learning*. New York, NY, USA: Springer, 2009.

[13] B. Schölkopf and A. J. Smola, *Learning With Kernels: Support Vector Machines, Regularization, Optimization and Beyond*. Cambridge, MA, USA: MIT Press, 2002.

[14] T. S. Furey, N. Cristianini, N. Duffy, D. W. Bednarski, M. Schummer, and D. Haussler, "Support vector machine classification and validation of cancer tissue samples using microarray expression data," *Bioinformatics*, vol. 16, no. 10, pp. 906–914, 2000.

[15] G. Guo, S. Z. Li, and K. L. Chan, "Support vector machines for face recognition," *Image Vis. Comput.*, vol. 19, no. 9–10, pp. 631–638, 2001.

[16] J. Taboada, J. M. Matías, C. Ordóñez, and P. J. García Nieto, "Creating a quality map of a slate deposit using support vector machines," *J. Comput. Appl. Math.*, vol. 204, no. 1, pp. 84–94, 2007.

[17] X. Li, D. Lord, Y. Zhang, and Y. Xie, "Predicting motor vehicle crashes using support vector machine models," *Accident Anal. Prev.*, vol. 40, pp. 1611–1618, 2008.

[18] T. Hansen and C. J. Wang, "Support vector based battery state of charge estimator," *J. Power Sources*, vol. 141, pp. 351–358, 2005.

[19] S. Haykin, *Neural Networks. A Comprehensive Foundation*. New York, NY, USA: Prentice Hall, 1999.

[20] J. Shawe-Taylor and N. Cristianini, *Kernel Methods for Pattern Analysis*. New York, NY, USA: Cambridge Univ. Press, 2004.

[21] S. Gun, M. Brown, and M. Bossley, "Network performance assessment for neurofuzzy data modeling," in *Proc. 2nd Int. Symp. Adv. Intell. Data Anal.*, 1997, vol. 1208, pp. 313–323.

[22] J. C. Viera, M. Gonzalez, B. Y. Liaw, F. J. Ferrero, J. C. Alvarez Anton, J. C. Campo, and C. Blanco, "Characterization of 109 Ah NiMH batteries charging with hydrogen sensing termination," *J. Power Sources*, vol. 171, pp. 1040–1045, 2007.

[23] C. Carballo, M. Gonzalez, J. C. Alvarez Antón, and C. Blanco, "A computerized system for testing batteries in full controlled environment," in *Proc. 17th IEEE Instrum. Measure. Technol. Conf.*, Gaithersburg, MD, USA, 2000, pp. 395–399.

[24] Electric vehicle battery test procedures manual, U.S. Advanced Battery Consortium (USABC), Southfield, MI, USA, 2006.

[25] T. Fletcher, "Support vector machines explained: Introductory course," University College London, London, U.K., Internal Rep., no. 0103, 2009.

[26] R. Picard and D. Cook, "Cross-validation of regression models," *J. Amer. Statist. Assoc.*, vol. 79, no. 387, pp. 575–583, 1984.

[27] L. Wasserman, *All of Statistics: A Concise Course in Statistical Inference*. New York, NY, USA: Springer, 2003.

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